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Beyond the Sick Role: The Many Roles of Adults With Type 1 and Type 2 Diabetes in the Management of Hypoglycemia—The InHypo-DM Study, Canada



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Key Messages

- Strategies and skills used in diabetes self-management have been reported, yet they have not been articulated as specific roles in hypoglycemia self-management.
- This qualitative study revealed that people with diabetes assume 4 roles to prevent or treat hypoglycemia: manager, technician, educator and advocate.
- Specific contexts (i.e. work, social settings, physical activity and travel) influence the patient roles.

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ABSTRACT

Objectives: Hypoglycemia is a common adverse event for people with type 1 and type 2 diabetes mellitus. In this article, we explore the specific roles that individuals assume to prevent or treat hypoglycemia.

Methods: A descriptive qualitative study from the Understanding the impact of HYPOglycemia on Diabetes Management Study (InHypo-DM) research program. A purposive sample of people with type 1 and type 2 diabetes were recruited for semistructured interviews. There were 16 participants (women and men), who were, on average, 53 years old. Average time since diagnosis was 15 (type 1) and 21 (type 2) years; all patients had at least 1 hypoglycemic event in the past year. Individual and team analysis of interviews were conducted to identify overarching themes.

Results: Participants articulated 4 roles in preventing or treating hypoglycemia. The first role was being a manager by assuming ownership and accountability for their own glycemic control. The second role, being a technician, used both subjective and objective information and employed specific strategies to respond to hypoglycemic events. The third role, educator, extended beyond self-management to increase others' awareness of hypoglycemia. The fourth role, advocate, involved championing one's own needs during a hypoglycemic event. These 4 roles were, in turn, influenced by the contexts of work, social settings, exercise and travel.

Conclusions: These findings demonstrate that strategies that individuals use to avoid or reduce the severity of a hypoglycemic event extend beyond merely making impromptu decisions during events. Instead, these 4 roles of manager, technician, educator and advocate, embedded in specific contexts, enhanced their mastery in managing hypoglycemia in daily life.

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R É S U M É

Objectifs : L'hypoglycémie est un effet indésirable fréquent chez les personnes atteintes de diabète sucré de type 1 et de type 2. Dans cet article, nous explorons les rôles spécifiques que les individus assument pour prévenir ou traiter l'hypoglycémie.

Méthodes : Une étude qualitative descriptive issue du programme de recherche sur l'Étude de Gestion du Diabète sur la Compréhension de l'incidence de l'HYPoglycémie (InHypo-DM). Un échantillon ciblé de personnes atteintes de diabète de type 1 et de type 2 a été recruté pour des entretiens semi-structurés. Il y avait 16 participants (femmes et hommes) qui avaient, en moyenne, 53 ans. Le temps moyen écoulé depuis le diagnostic était de 15 ans (type 1) et 21 ans (type 2); tous les patients ont eu au moins un événement hypoglycémique au cours de l'année écoulée. Les analyses des entretiens ont été menées individuellement et en équipe afin d'identifier les thèmes prédominants.

Résultats : Les participants ont articulé 4 fonctions dans la prévention ou le traitement de l'hypoglycémie. La première fonction était d'être un gestionnaire en assumant la pleine propriété et la responsabilité de leur propre contrôle glycémique. La deuxième fonction, celle de technicien, consistait à utiliser des informations à la fois subjectives et objectives et à employer des stratégies spécifiques pour répondre aux événements hypoglycémiques. La troisième fonction, celle d'éducateur, allait au-delà de l'autogestion, afin d'accroître la sensibilisation des autres à l'hypoglycémie. La quatrième fonction, celle d'avocat, consistait à défendre ses propres besoins lors d'un événement hypoglycémique. Ces quatre fonctions étaient à leur tour influencées par les contextes de l'environnement de travail, des milieux sociaux, de l'exercice et des voyages.

Conclusions : Ces résultats démontrent que les stratégies utilisées par les individus pour éviter ou réduire la gravité d'un événement hypoglycémique vont au-delà de la simple prise de décisions imprévisibles lors d'un événement. Au contraire, ces 4 fonctions de gestionnaire, de technicien, d'éducateur et d'avocat, ancrés dans des contextes spécifiques, ont renforcé leur maîtrise dans la gestion de l'hypoglycémie dans la vie de tous les jours.

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Introduction

In this study, we explore the experience of hypoglycemia in people living with diabetes and how this impacts their lives, including the roles that they assume when managing their risk of hypoglycemia. The concept of the “sick role” was introduced by sociologist Talcott Parsons in 1951. He argued that the individual granted “sick role” status by society was afforded certain rights, such as the right to be excused from work and family obligations, but in return must continually strive to get well and must comply with the recommendations of medical experts (1). Medical sociology has since then moved beyond this concept of placing the patient in a “passive, dependent role” (2). However, the role(s) patients assume when managing chronic diseases, such as diabetes mellitus, specifically in relation to hypoglycemia, has not yet been conceptualized in the literature. Therefore, it is important to ascertain how this patient population enacts various evolving roles that support self-management.

Chronic disease can alter multiple layers of a person's life and sense of self (3,4). In the case of diabetes, this requires significant changes to one's lifestyle on multiple levels, such as diet, exercise, medication regimens, monitoring of blood glucose and management of hypoglycemia (5). The use of insulin and/or secretagogues can result in hypoglycemia, a clinical syndrome characterized by abnormally low blood glucose (6). Recently, the UnderstandINg the impact of HYPoglycemia on Diabetes Management Study (InHypo-DM), Canada, using a self-reported questionnaire, demonstrated that the population burden of severe hypoglycemia among people with diabetes is substantial in both type 1 diabetes (2.5 events per person year) and type 2 diabetes (2.4 events per person year) (7). Moreover, the burden of nonsevere hypoglycemia was even higher at 55.7 events per person year in participants with type 1 diabetes and 28.0 events per person year in participants with type 2 diabetes (7).

Neurogenic symptoms of hypoglycemia include anxiety, sweating, tremor, hunger and pallor, whereas neuroglycopenic symptoms include behavioural changes, cognitive impairment, seizure and coma (8). The risk and experience of these symptoms has been connected to a range of emotional responses, including frustration, anxiety and concerns about safety (9,10).

Fortunately, many people with diabetes engage in positive strategies to prevent and manage hypoglycemia (11,12). The diabetes self-management literature indicates that, to successfully manage diabetes and hypoglycemia, the individual needs to: have developed motivation, which may be influenced by their health beliefs; have opportunities for acquiring health literacy; and have developed self-efficacy and problem-solving skills (13–15). These elements have been articulated “in terms of the patient” in the literature but not specifically applied to the treatment of hypoglycemia.

Although previous studies have reported various strategies to support self-management of diabetes, we describe how these strategies translate into specific roles in the self-management of hypoglycemia.

Methods

This descriptive qualitative study is part of the InHypo-DM mixed methods study conducted in Canada (7,16–18). The analysis draws upon the qualitative patient data, which explored the experiences of hypoglycemic events among people with diabetes.

Participant recruitment

We used a purposive sampling technique to recruit participants, with posters placed in various locations (e.g. local diabetes clinic, family doctors' office, pharmacies, shopping malls) throughout southwestern Ontario. We sought to attain a maximum variation

sample (age, sex, type of diabetes, age of onset). Participants had to be on insulin and/or a secretagogue and to have had at least 1 hypoglycemic event in the previous 12 months. The health sciences research ethics board of Western University approved this study. Participants were recruited separately from the sample recruited for the InHypo-DM quantitative study.

Data collection

A 30- to 45-minute semistructured interview was conducted with each participant. An example of an interview question was: “What role do you play in preventing a hypoglycemic event?” Informed consent was obtained before the interview began. The interviews were conducted by a member of the research team at a location convenient for the participant. All interviews were audiotaped and transcribed verbatim. Data collection and analysis took place concurrently. Therefore, data collection ceased upon reaching saturation in that there were no new themes emerging.

Data analysis

The data analysis was both iterative and interpretative using individual and team analysis. In the initial phase of the analysis, 4 members of the research team independently reviewed each transcript to identify key concepts emerging from the data. The team then met to compare the independent reviews, culminating in the development of the coding template, which evolved over the course of the analysis. NVivo 10 software was used for coding and organizing the data. Upon completion of this phase, a subset of the team met to further synthesize and interpret the main themes. Throughout the analysis, the research team identified exemplar quotes reflecting the main themes. In the Results section, the exemplar quotes are identified by type of diabetes (e.g. T1), sex of participant (e.g. F for female) and participant number.

Credibility and trustworthiness

Credibility and trustworthiness of the data were enhanced through verbatim transcripts, field notes generated after each interview and independent and team analysis. Reflexivity was essential during the analysis. The authors come from different backgrounds (e.g. social work, family medicine, epidemiology), so we reflected on the ways in which our own values and experiences shaped interpretation and reporting of the data.

Final sample

The final sample consisted of 16 participants, which included 10 women and 6 men with type 1 (8) and type 2 (8) diabetes. The average age was 40 (range, 22 to 64) years in the type 1 diabetes group and 60 (range, 45 to 77) years in the type 2 diabetes group. The time since diagnosis was an average of 15 years in the type 1 diabetes group and 21 years in the type 2 diabetes group. Fourteen of the 16 participants had drug coverage, either through provincial or third-party insurance. All had had at least 1 hypoglycemic event in the past year, ranging from severe to nonsevere.

Results

Our analysis revealed how participants assumed 4 main roles in preventing or treating hypoglycemia. The first 2 roles were manager and technician, employing a variety of strategies to prevent hypoglycemia. The third and fourth roles, educator and advocate, extended beyond self-management to involve others. These 4 roles were in turn influenced by different contexts including work, social settings, physical activity and travel (Figure 1).

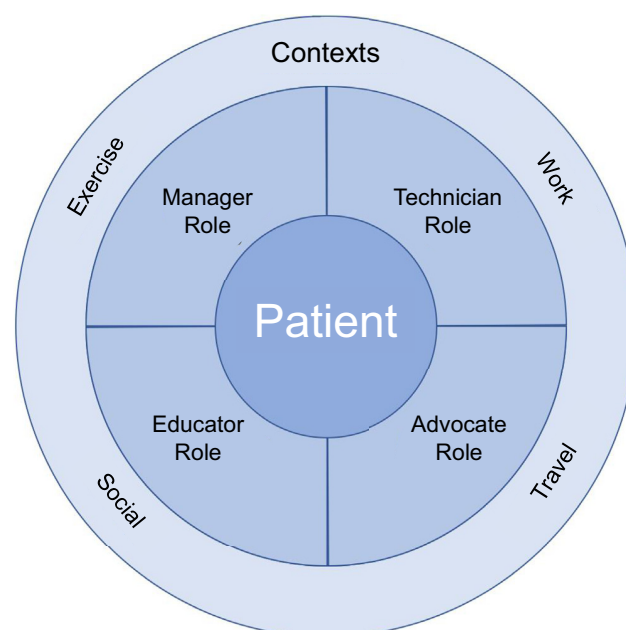


Figure 1. The contexts and roles of patients with diabetes in the prevention and management of hypoglycemia.

Manager role

Participants strongly believed it was their responsibility to become the manager or overseer of their diabetes and hypoglycemia: “My role? If I don’t manage it, nobody else is going to” (T2M-7). This included taking ownership and being accountable for their glycemic control: “Just keep it in control and be prepared” (T2F-4). In the manager role, participants described coming to the realization that they could not rely on external supports, prompting them to take responsibility for themselves: “I play sports. I have lollipops in my bag ... There’s also a vending machine that is there. Now, it has been out of order on the odd occasion ... Don’t rely on it ... Take responsibility for yourself because those resources that you think are going to be there may not be there” (T2F-4).

Anticipating that hypoglycemic events were sometimes unavoidable further encouraged this aspect of being a manager: “I think you always need to be prepared for low blood sugars in the event that they happen ... people who think that they’re never going to happen will get caught not prepared” (T2F-4).

Technician role

Diabetes mellitus is unique among chronic illnesses in that self-management requires highly technical skills. A technician uses both subjective and objective information to respond to hypoglycemic events. This involves identifying cause and effect, measuring changes and adjusting doses accordingly. Participants described how they had become attuned to the early signs and symptoms of hypoglycemia: “Any time that I notice a difference. Things like perspiration or energy levels or... dizziness. Those are the things that make me carefully approach the issue and check my sugar right away” (T2M-6).

Symptom awareness prompted some participants to examine the cause and effect precipitating the hypoglycemic event: “I think my role is knowing the symptoms and being aware of them and ... looking at what caused them and how to alleviate that in the future” (T1F-4).

Objective information was obtained through frequent testing of their blood sugar: “I test all the time. I test before I go driving. I test

before I go for a walk ... then if it's low or starting to go low, then I'll take something" (T2F-5). As technicians, participants described the importance of combining both subjective and objective sources of information: "I monitor very closely, especially if I'm starting to get a lower reading ... then I'm really aware if there's any change in the way I feel" (T1M-2).

In addition, the technician role included the ability to process the information they were receiving and adjusting their treatment protocol accordingly: "I usually reduce my insulin if I'm going to become active, try to make sure I count my carbohydrates accurately so I'm not giving any more insulin than needed" (T1F-4). This often involved a complex balancing act between medication, diet and exercise: "So 4 times a day I do my actual blood glucose and I'm calibrating the monitor and (if) symptomatic I still do a blood glucose. Just trying to eat well ... Looking at the glycemic index and how it works with my symptoms" (T1F-3).

Educator role

Although the previous 2 roles of manager and technician were universal among participants, the roles of educator and advocate were more ad hoc, often in response to a specific situation. As educators, participants involved others in the management of their hypoglycemia by explaining what hypoglycemia is and what to do if they had an event: "I tell all my classmates I'm a diabetic. This is what happens and I let them know that if they can't get sugar in my mouth, then just call the ambulance" (T1F-5).

Participants acknowledged that self-sufficiency was not always realistic and they needed to involve others: "(If) you don't admit to people you have diabetes and you do go low now and then, how are your friends going to help you?" (T2F-4).

Advocate role

A defining feature of the advocate role was being active in the moment of dealing with hypoglycemia. One participant described both positive and negative experiences in self-advocacy. On the positive side: "...most people are pretty understanding when you tell them I've got low blood sugar ... (if) I need somebody to grab me a pop, most people would be like, 'okay, I'm on it'" (T1M-3).

In contrast, the same participant described a more negative side of needing to be one's own advocate: "I've had like run-ins where people are like, 'you can't do that' (and) I'm like, 'stop me.' I've got low blood sugar, you want me to pass out on the floor versus, you know, getting some sugar into me" (T1M-3).

Participants tried not to inconvenience others, whereas still working to champion their own needs and safety: "I've learned from past mistakes. One time at the gym ... I could have sworn I always had a fast-acting carb, and in this case I didn't, and that's when I needed it. I just remember asking a gym employee ... And I felt embarrassed when I knew I shouldn't" (T1F-2).

Contextual influence on roles

Participants described how specific contexts could influence which role they embodied. These contexts included work, social settings, physical activity and travel.

Work: Some work situations required participants to be prepared for a potential hypoglycemic event by assuming the roles of both manager and technician. Participants needed to be aware of their symptoms at work and manage them professionally: "I think at work sometimes if I'm seeing a patient and I can feel myself going low, then I will come up with some excuse that I have to leave the room" (T1F-4).

Furthermore, during a hypoglycemic event, the role of the technician would become a top priority at work: "If I am at work,

you literally have to stop and treat. Everything kind of goes on hold until I can get (my blood sugar) back up" (T1F-3).

Other work situations required that they assume the educator role and alert their colleagues to a change in their status. In fast-paced work environments, educating coworkers was essential, although, at times, uncomfortable: "I've had a couple (of hypoglycemic events) at work. And at first you feel kind of embarrassed because you just need to take a time out. And work is kind of quick-paced ... But I had let the team know that I had just started out on a pump and I was figuring out my values" (T1F-2).

Social settings: Attending social activities could place participants at risk for hypoglycemia, requiring them to be both a manager and a technician: "If I'm drinking at a party then I check that (blood sugar) more often and make sure to have snacks" (T1F-2). Although some participants expressed embarrassment in asking for help in a social situation, they also recognized that they should advocate for their needs: "I do play a lot of bridge, so I've neglected to ask the hostess for something when I start getting the shakes ... And I think that's a mistake. I have to start asking for a cookie or something ... I shouldn't be afraid of that" (T2F-3).

Physical activity: For several participants, exercise was a central component of adopting a healthy lifestyle. The following quote reveals the role of manager and the strategies used to be prepared: "If I'm going out to play golf ... I generally try to load up on carbs before I hit the course, which really does seem to help. I also have a golf bag full of fast hit treats in case there's a problem" (T1M-2).

When participants engaged in strenuous activity, the technician role was critical in performing the balancing act between medication, diet and exercise: "So if you want to be exercising you prep ahead of time, so take a snack and I program a temp basal so I can reduce the amount of insulin my pump is giving for a period of time" (T1F-3).

Travel: Travel presented unique challenges requiring participants to be a manager and plan accordingly: "If I am traveling, just getting extra things for the trip in case I lose something" (T1F-2). Having additional supplies on hand ensured they could adequately enact their technician role: "Especially when I travel, I make sure I have my test kit with me and I've got enough strips ... they're up to date and that sort of thing" (T2F-4).

Discussion

In this article, we have captured 4 specific roles assumed by people living with diabetes to prevent or manage hypoglycemia. As a manager, they take ownership and responsibility for their own glycemic control. As a technician, they identify cause and effect, measure changes and adjust doses accordingly. As an educator, they act as an ambassador for diabetes to the people around them, explaining what hypoglycemia is and what to do if they experience an event. As an advocate, they champion their own needs in the moment. In different contexts, they must attempt to balance competing spheres of their lives and assume a role fitting to each setting. These roles were equally reflected by our participants with both type 1 and type 2 diabetes.

Findings illuminate 4 specific roles that participants assumed. Some evidence of the attributes of these roles exists in the literature, without being named as such. The manager and technician roles have often been discussed indirectly in the chronic disease self-management literature (19). The concept that self-management encompasses multiple responsibilities was well defined by Moser et al (11), who noted that self-management is a process that incorporates "daily," "off-course" and "preventative" processes. The roles of manager and technician are observed in the

steps required to complete these processes, including: adhering to medications, becoming aware through monitoring and listening to their body and deciding on a course of action (11).

There has been a notable change in attitude toward discussing hypoglycemia, as reflected by our participants' eagerness to assume the role of educator, whether it be with friends, family or coworkers. This finding is in sharp contrast to research on living with hypoglycemia from 20 years ago, where participants described a hypoglycemic event as a "private experience" (20). Our findings are more concordant with Paterson and Thorne's (2000) (21) evolution of self-management expertise as, in their findings, expertise was achieved when "participants learned to assist others to become the partners they needed in their management" (p 414). Our study participants highlighted the importance of their role as educators in ensuring those around them were aware of how to respond to a hypoglycemic event.

With regard to the advocate role, one context in which people with diabetes are known to act as self-advocates is in clinical encounters with health-care providers (HCPs). The role of the patient in both providing information and articulating their own preferences was described in the shared decision-making model conceptualized by Charles et al (1997) (2). Patient self-advocacy in the clinical encounter is often necessary because of paternalistic attitudes held by HCPs (22). In comparison, the context in which our participants most often described self-advocacy was not in interactions with HCPs, but in interactions with friends, coworkers and the general public. This is a new finding not described in the extant literature.

Participants described how the contexts of work, social settings, physical activity and travel could influence which role was enacted. The roles and strategies employed by participants in the workplace context can potentially mitigate the challenges of dealing with hypoglycemia events at work, as cited previously in the literature (23,24). Earlier research has addressed the reluctance of people with diabetes to have a hypoglycemic event witnessed by others or to seek help in its management (20). In contrast, our study participants expressed a desire to assume the educator and advocate roles in social settings. However, consistent with earlier research, this was dependent on their ability to overcome feelings of embarrassment and social discomfort (20).

For the majority of participants, physical activity is an important part of their lives. Fear of hypoglycemia while participating in physical activity, often cited as a barrier (25), did not appear to prevent participants from engaging in exercise. Similarly, concerns about hypoglycemia did not appear to impede participants' pursuing travel. Rather, they had adopted many of the recommendations for avoiding hypoglycemia during travel, such as having a supply of fast-acting carbohydrates and adequate medical supplies as well as planning ahead (26). In addition, the rapid advancements in new technologies, such as continuous glucose monitoring, more accurate meters and smart technology, with downloading and connectivity functions, increasingly facilitates patients' ability to travel more freely and with less worry.

What our findings add to the literature is the precise identification and definition of the roles assumed by people with diabetes in the self-management of hypoglycemia. These roles were universal across our findings—participants articulated the defining behaviour, if not necessarily the name, of each role. How then does this relate to Talcott Parsons's antiquated concept of the sick role? To some extent, these roles are an interpretation of how the sick role can be manipulated to live the fullest life possible. In some contexts, a person with diabetes at risk of hypoglycemia may embrace or derive power from self-identifying as a sick person, such as when acting as an educator. At other times, they defy the sick role's limitations, when advocating for themselves or participating in work, social activities, exercise and travel.

One limitation of this study is that the data were collected at only 1 time-point in the participants' lived experiences with diabetes. The Shifting Perspectives model of chronic illness describes the patient's focus as shifting between "illness in the foreground" vs "wellness in the foreground" (27). Depending on which of these 2 states participants are in at the time of the interview, they may give starkly different answers. Therefore, future longitudinal research could explore each participant's changing perspectives across time. We did not identify a difference between women and men in their description or enactment of the 4 roles. This is in contrast to earlier research describing the different challenges and expectations of women and men in their diabetes self-management (28). Future research is required on the roles assumed by men and women in diabetes self-management—particularly hypoglycemia—to provide gender-appropriate treatment and education. In interpreting the findings, we were limited by the fact that we did not have data on the type and duration of medication use, and this should be considered in future studies. By recruiting participants from various locations such as shopping malls and physician offices across the region of southwestern Ontario, we attempted to investigate a diverse group of participants. Accordingly, similar to potential gender differences, future research should address a broad range of sociodemographic characteristics in order to develop treatment and education that is appropriate for people of all backgrounds.

In conclusion, these findings demonstrate how strategies that people with diabetes use to avoid or reduce the severity of a hypoglycemic event extend beyond merely making impromptu decisions during events. Instead, these well-delineated roles of manager, technician, educator and advocate, embedded in specific contexts, highlight the abilities and confidence of people with diabetes in managing hypoglycemia. Each role was characterized by specific actions and responsibilities.

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Author Contributions

J.B.B. was involved in conception and design of the study, all levels of analysis and interpretation of the data; drafting and reviewing the article; and final approval of the version to be submitted. Y.V. was involved in analysis and interpretation of the data, drafting and reviewing the article and final approval of the version to be submitted. C.M. was involved in analysis and interpretation of the data, drafting and reviewing the article and final approval of the version to be submitted. S.M.R. was involved in conception and design of the study and preliminary analysis, revising the article critically for important intellectual content and final approval of the version to be submitted. S.W.B. was involved in the conception and design of the study and preliminary analysis, revising the article critically for important intellectual content and final approval of the version to be submitted. A.R.L. was involved in conception and design of the study and preliminary analysis, revising the article critically for relevant intellectual content and final approval of the version to be submitted. B.L.R. was involved in conception and design of the study, revising the article critically for important intellectual content and final approval of the version to be submitted. S.H. was involved in conception and design of the study, revising the article critically for important intellectual content and final approval of the version to be submitted.

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